

Topics of today's class

Last class:

- Chapter 5
 - A Closed Economy One-Period Macroeconomic Model

Today's class:

- Chapter 7
 - Economic growth facts
 - Malthusian model of economic growth

Chapter 7

Economic Growth: Malthus and Solow

Chapter 6 Topics

- Economic growth facts
- Malthusian model of economic growth
- Solow growth model
- Growth accounting

Growth: Why should we care?

In which country:

- life expectancy at birth is under 50 years
- 1 out of 10 infants dies before reaching a first birthday
- 90 percent of households have no electricity
- less than 10 percent of young adults have graduated from high school

Growth: Why should we care?

In which country:

- life expectancy at birth is under 50 years
 - 1 out of 10 infants dies before reaching a first birthday
 - 90 percent of households have no electricity
 - less than 10 percent of young adults have graduated from high school
-
- It is the U.S. in 1900.

There are large variations over time in a country standards of living

- GDP per capita in U.S. was about \$2800 in 1870
- rose to more than \$42,000 by 2009, almost a 15-fold increase.

Growth: Why should we care?

There are large variations across countries in standards of living.

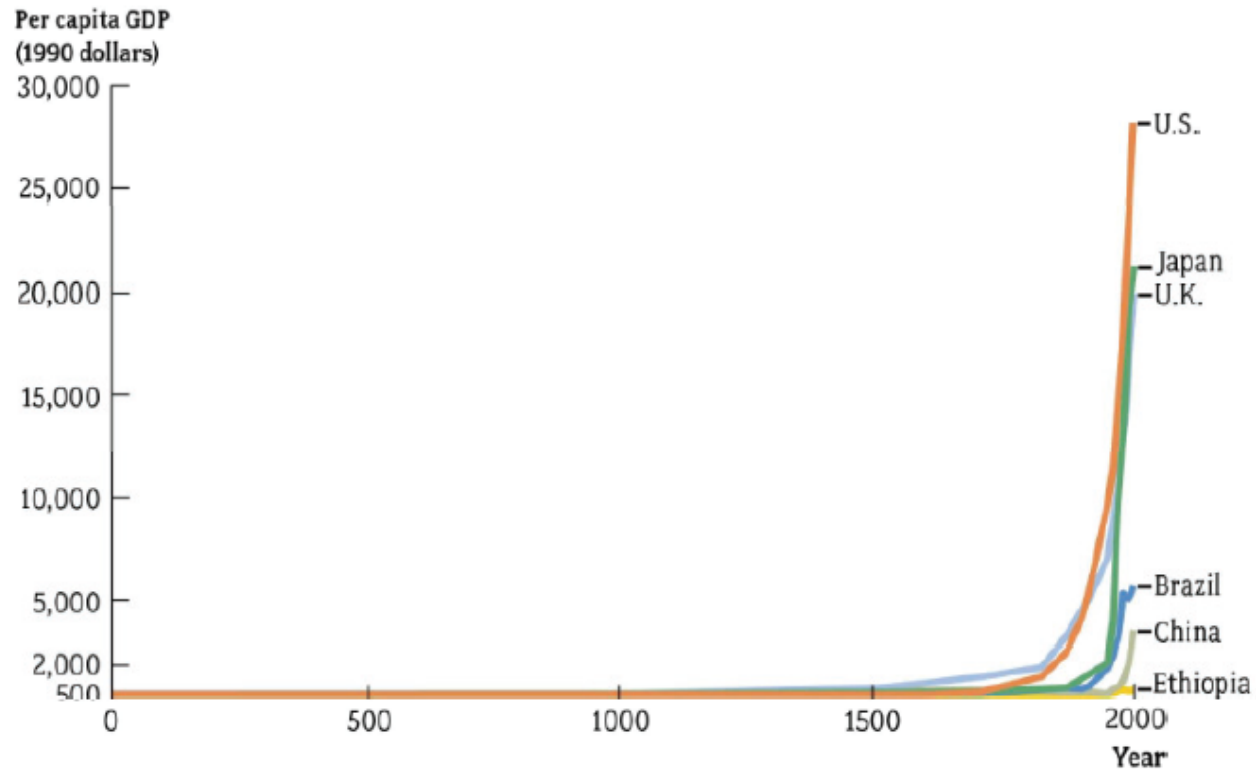
- Ratio of GDP per capita USA/Congo: 91
- Life expectancy in rich countries is 77 years, 67 years in middle income countries, and 53 years in poor countries.
- Out of 6.4 billion people in the world, 0.8 do not have access to enough food, 1 to safe drinking water, and 2.4 to sanitation.

Growth: Why should we care?

- Economic Growth is correlated with long-run standards of living.
- Striking example:
 - Real GDP per capita in Argentina and Canada were the same in 1900: \$2,750.
 - Argentina's average growth rate in 1900-2009 was around 1.605%. Canada's average growth rate in 1900-2009 was around 2.463%.
 - In 2009 the per capita GDP of Canada was \$39,000, while Argentina's GDP was \$16,000.

Growth: Why should we care?

Economic Growth over the Very Long Run in Six Countries



Source: Angus Maddison, *The World Economy: Historical Statistics* (Paris: OECD Development Center, 2003).

Economic Growth Facts

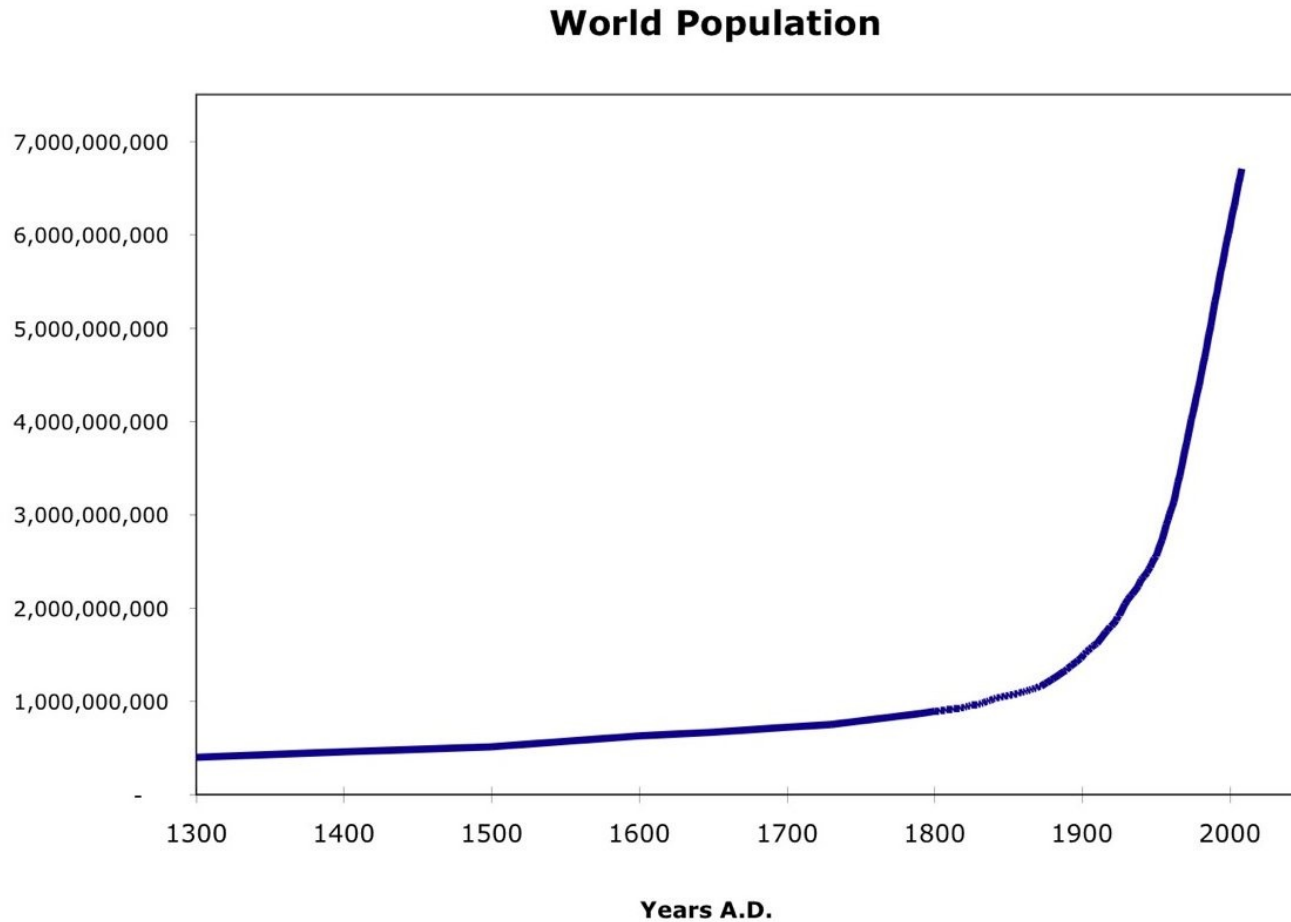
- Growth is remarkably a recent phenomenon.
- Before the Industrial Revolution around 1800, standards of living differed little over time and across countries.
- Since the Industrial Revolution, per capita income growth has been sustained in the richest countries.

World Population and GDP

Year	Population*	GDP per Capita**
-5000	5	\$130
-1000	50	\$160
1	170	\$135
1000	265	\$165
1500	425	\$175
1800	900	\$250
1900	1625	\$850
1950	2515	\$2030
1975	4080	\$4640
2000	6120	\$8175

(Population is in millions and GDP per Capita is in 2000 \$)

World Population



Malthus

- Thomas Malthus: An essay on the Principle of Population (1798).
- Population growth has a negative impact on the standard of living, because increases in population dilute any gains in productivity.
- He argues in favor of population control or “forced family planning”.

Malthusian model of economic growth

- Predicts that a technological advance will only increase population, with no long-run change in the standard of living.

Production Function

Output is produced from land and labor inputs.

$$Y = zF(L, N)$$

- L: land instead of capital
- Y: food

Dynamic Model

- before: static models
- now: dynamic model
- no-investment: there is no way to store food from one period to the other.

Evolution of the population

- $N' = N + \text{Births} - \text{Deaths}$
- $N' = N + N \times (\text{birth rate} - \text{death rate})$

Evolution of the Population

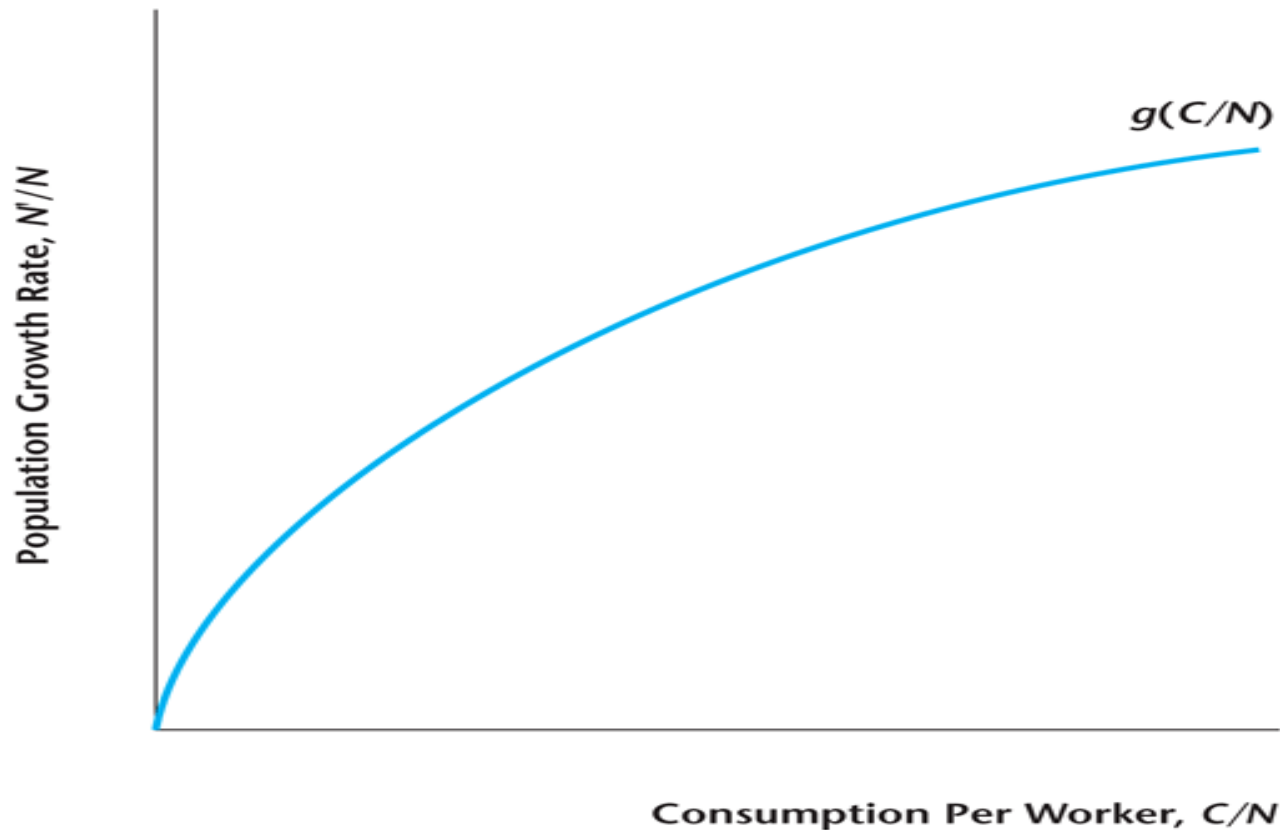
- Higher population growth, the higher is per-capita consumption.

$$N' / N = g(C / N)$$

- Birth rate increases with consumption per capita
 - people will have more children by choice.
 - better nutrition increases fertility.
- Death rate decreases with consumption per capita
 - decreases infant mortality.
 - population healthier, increasing the average lifespan.

Figure 6.5

Population Growth Depends on Consumption per Worker in the Malthusian Model



Equilibrium Condition

- In equilibrium, consumption equals output produced.

$$C = zF(L, N)$$

income expenditure identity for this economy

Equilibrium Evolution of the Population

- This equation describes how the future population depends on current population.

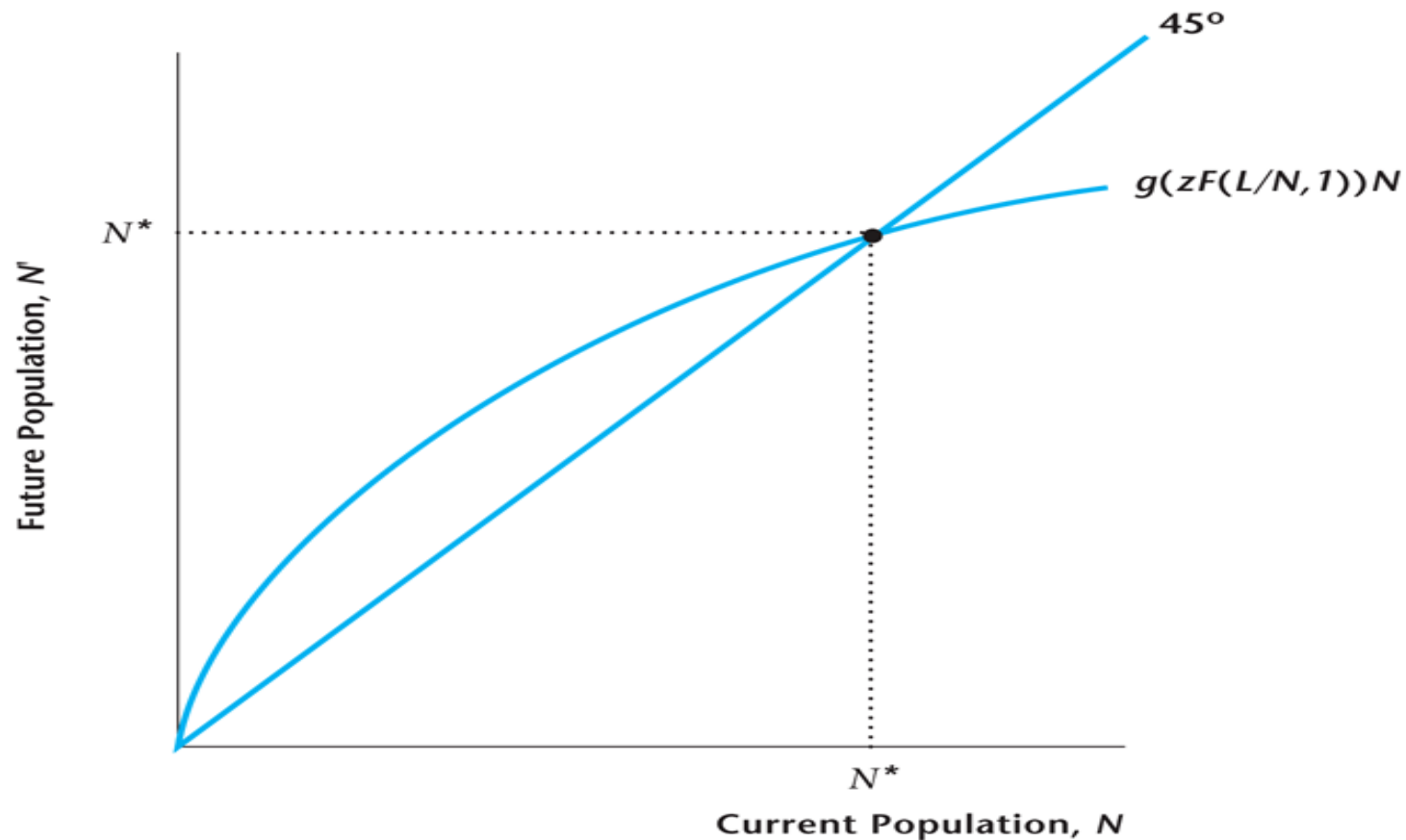
$$N' / N = g[zF(L, N) / N]$$

How Population Evolves in Equilibrium

$$N' = g \left[z^F \left(\frac{L}{N}, 1 \right) \right] N.$$

Figure 6.6

Determination of the Population in the Steady State



Steady state of the model

$$Y / N = zF(L / N, 1)$$

or

$$y = zf(l)$$

where y : output per worker, l : land per worker

- Since consumption equals output, consumption per worker c is:

$$c = zf(l)$$

A Steady State Condition

Population growth is increasing in consumption per worker, c

$$N' / N = g(c)$$

Figure 6.7

The Per-Worker Production Function

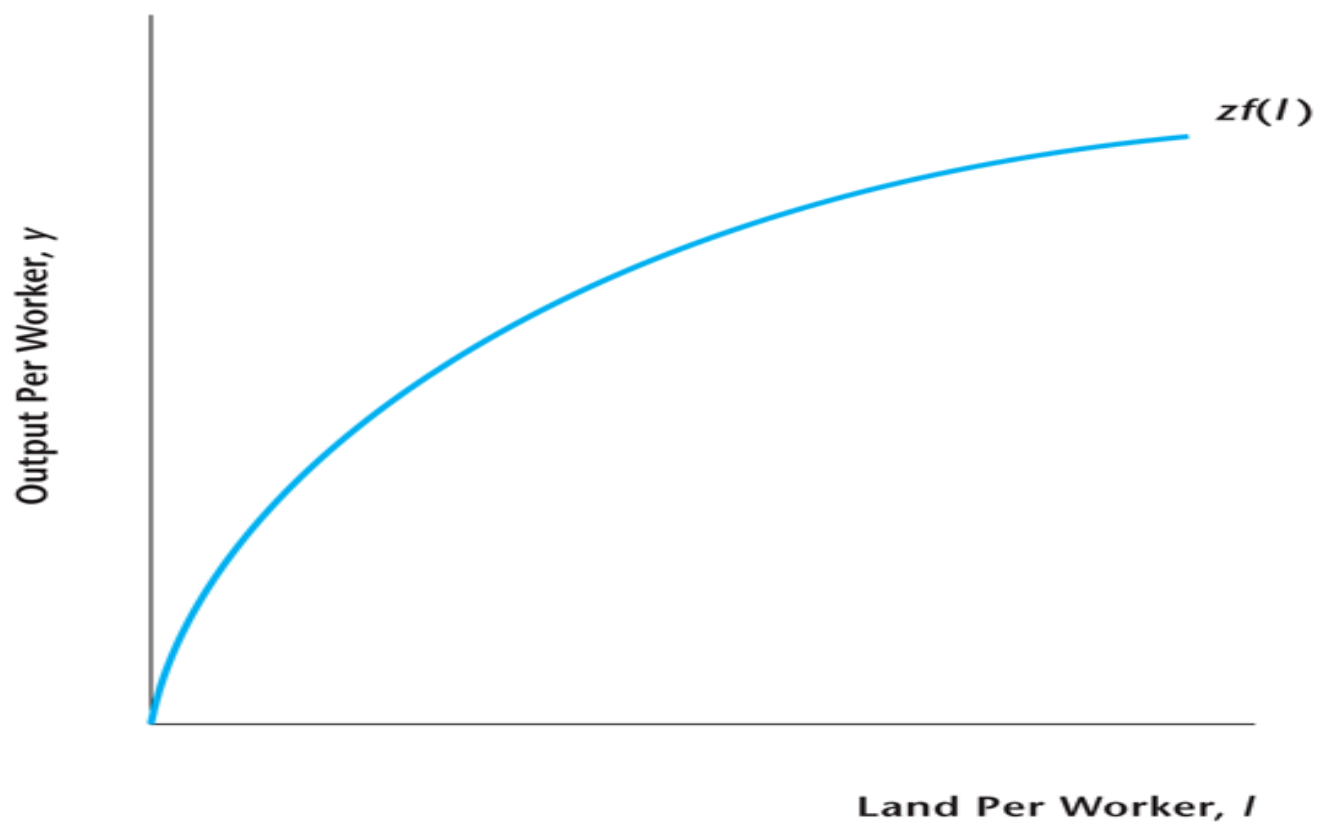
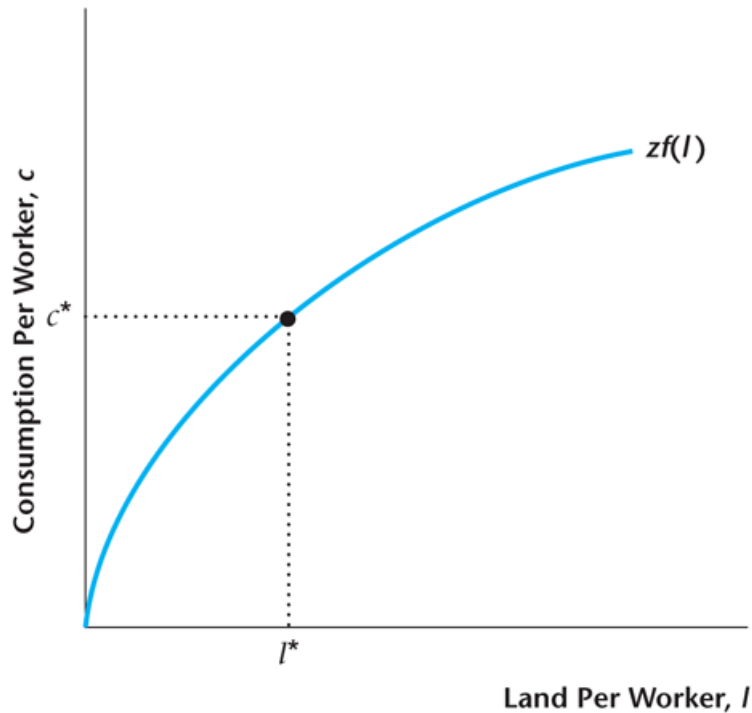
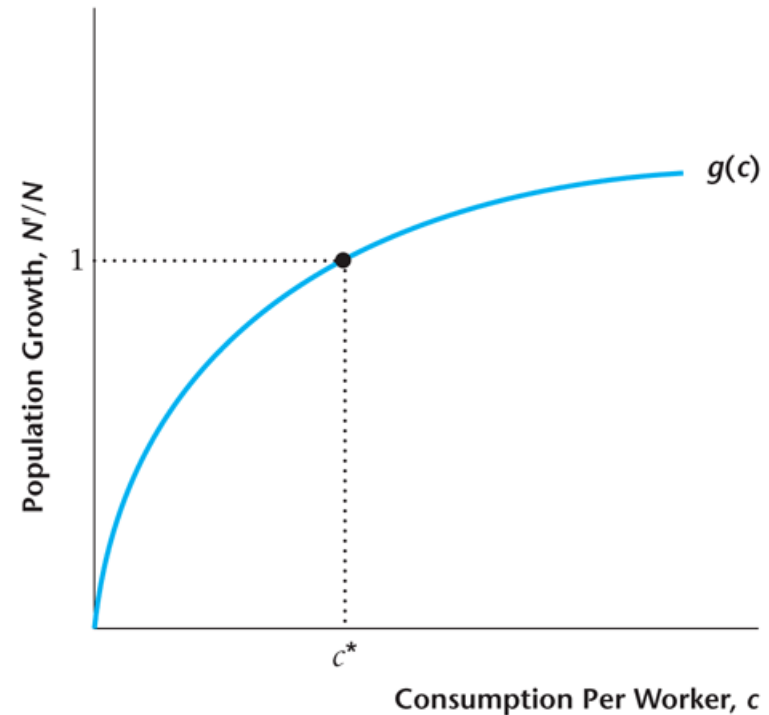


Figure 6.8

Determination of the Steady State in the Malthusian Model



(a)



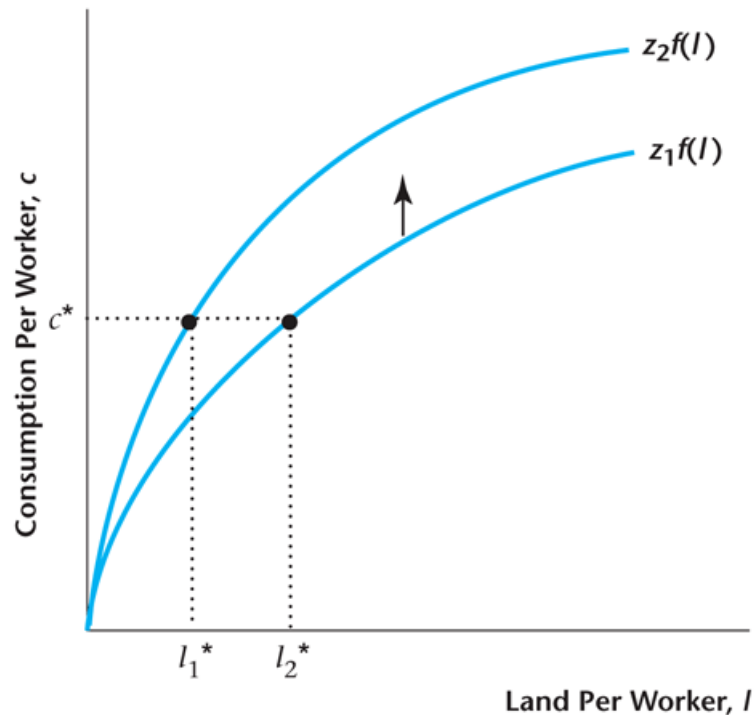
(b)

An Increase in z in the Malthusian Model

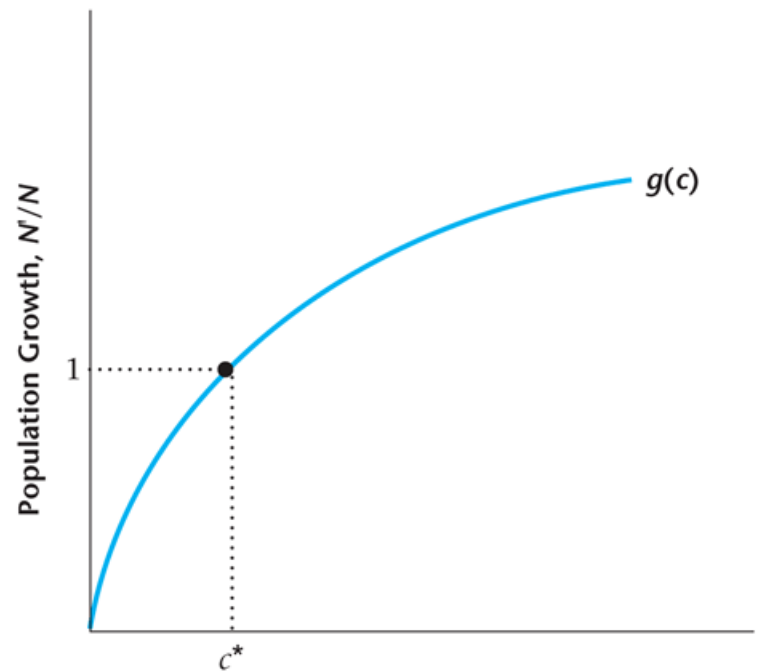
- If z increases, this shifts up the per-worker production function.
- In the long run, the population increases to the point where per capita consumption returns to its initial level.
- There is no long-run change in living standards.

Figure 6.9

The Effect of an Increase in z in the Malthusian Model



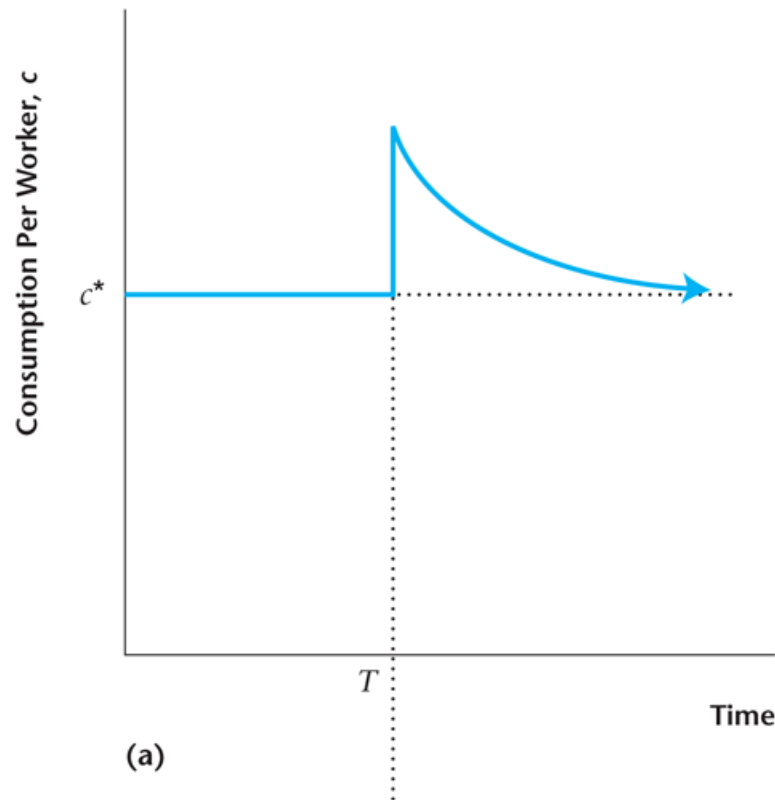
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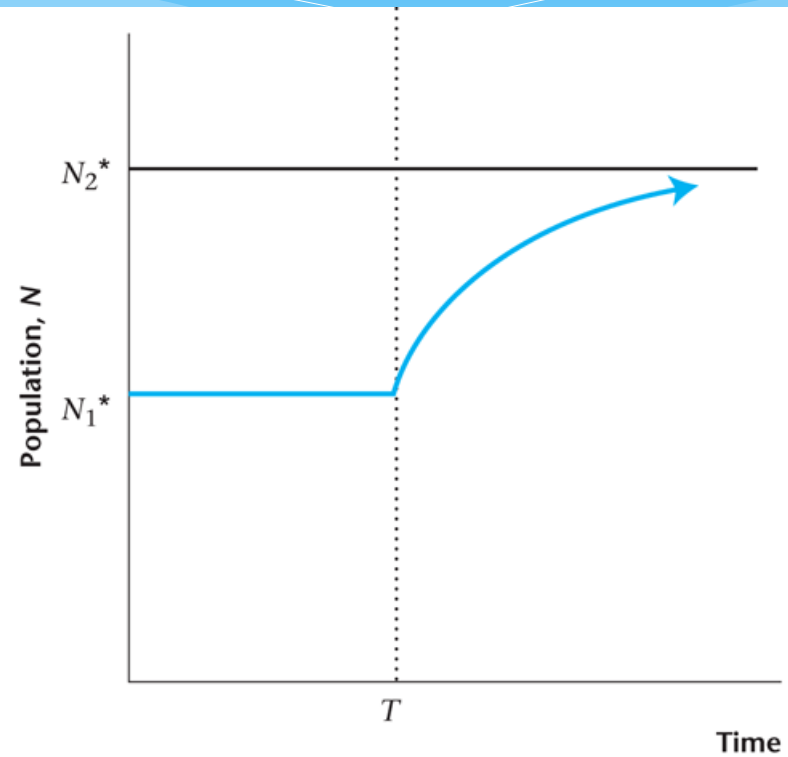
(b)

Figure 6.10

Adjustment to the Steady State in the Malthusian Model when z Increases



(a)



(b)

An increase in TFP

Immediate Effect of an Increase in TFP

- increase in the productivity of labor
- and so, an increase in consumption per capita

But increase in consumption per capita leads to:

- an increase in population size
- a decrease in the productivity of labor (decreasing marginal return to labor)

In the long-run:

- consumption per worker is unchanged
- population increases

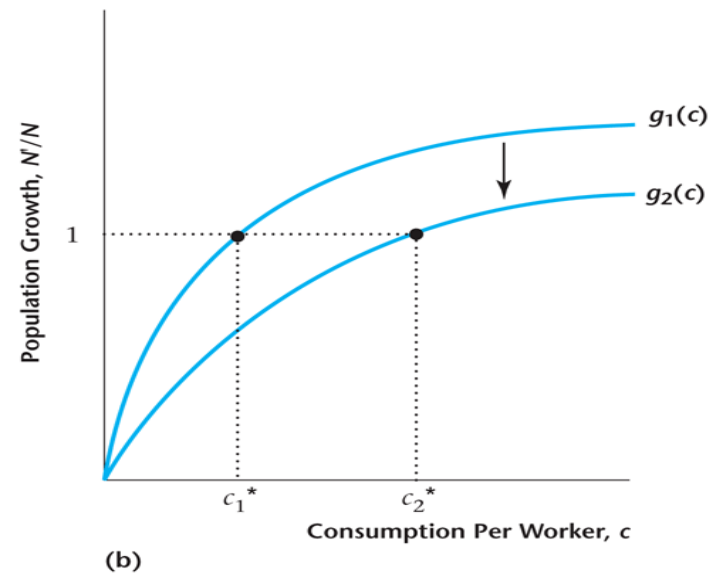
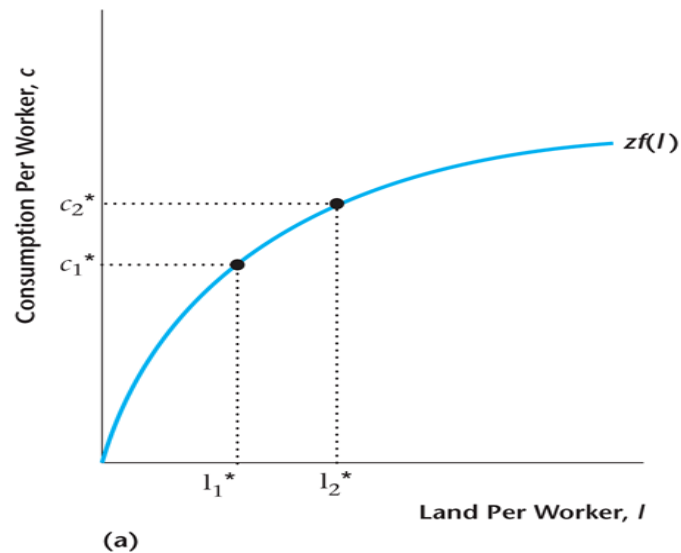
Pessimistic View: improvements in technology do not improve standard of living in the long run.

Population Control in the Malthusian Model

- Population control alters the relationship between population growth and per-capita consumption.
- In the long run, per capita consumption increases, and living standards rise.

Figure 6.11a

Population Control in the Malthusian Model



How Useful is the Malthusian Model?

- Model provides a good explanation for pre-1800 growth facts in the world.
- Malthus did not predict the effects of technological advances on fertility.
- Malthus did not understand the role of capital accumulation in growth.